AI-based cotton disease detection and advisory system using a convolutional neural network (CNN).

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Abstract - **Cotton is essential cash crop because most farmers grow it in huge quantities. However, many diseases reduce the quality and quantity of cotton crops, resulting in a significant loss in production and productivity. It is critical to detect these disorders at an early stage of diagnosis. In order to solve the above problem, an intelligent detection and classification system for cotton diseases based on the image of plants is proposed, which is called CottonCare AI. The proposed system is based on image classification approach using a Convolutional Neural Network (CNN) to analyse the images of the cotton plant and classify the condition. The labelled dataset is used to develop the model and contains 8 classes, Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot. Various CNN architectures such as MobileNetV2, EfficientNetB0 and ResNet50 are trained and evaluated for finding an appropriate model for the application. Of these, MobileNetV2 had an accuracy of 95.23% and an approximate model size of 9.30 MB and is suitable for efficient image-based prediction. The detected condition is displayed for the user with its confidence level, symptoms, suggested solution, and prevention tips. The system also includes a prediction history and an advisory feature using a chatbot to enhance the usability for farmers. CottonCare AI is designed to offer an easy and smart solution for early cotton disease detection and help farmers make informed decisions on crop management.**

Keywords: CNN, Deep Learning, MobileNetV2, Transfer Learning, Image Classification, TensorFlow Lite, Precision Agriculture.

I. INTRODUCTION

Agriculture is one of the key industries of Indian economy and provides livelihood to a large population. Agriculture is an important crop in India and provides livelihood for millions of people, besides making an important contribution towards textile industry. cotton yield approximately reaching 297.24 lakh bales. Cotton production in India takes place mainly in central and southern regions of the country. The main cotton producing states are Maharashtra, Gujarat, Telangana, Karnataka, Rajasthan, Madhya Pradesh, Andhra Pradesh and Haryana. Leading among the states in terms of area is Maharashtra which produced around 73.73 lakh bales, followed by Gujarat and Telangana with cotton production being approximately 71.57 lakh and 57.89 lakh bales respectively. Despite being an economically important crop, cotton production is affected by various diseases and insect pests. Examples are aphids, army worms, bacterial blight, cotton boll rot, powdery mildew etc. can cause plant growth reductions, poor quality, and decreased production. Therefore, early identification is crucial in the effort to control the spread of disease and minimize damage. Farmers have historically learned about these issues from visual inspection, from their experience or from a specialist. Manual identification is time consuming, however, and can lead to misdiagnosis, particularly because of the similarities in disease symptoms.

AI, image processing, and deep learning are among the latest technologies that are proving to be valuable solutions for automated plant disease detection. To get plant visual features directly from plant images, Convolutional Neural Networks (CNNs) can be trained.Convolutional Neural Networks (CNNs) are able to be trained to learn features from plant images, including color, texture, shape and patterns. CottonCare AI is designed to identify and categorize cotton diseases and pests using the images of plants.

The proposed system has a labelled dataset with the following 8 classes: Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot. These models are trained and tested, and MobileNetV2 is found to be the most light weight with accuracy of 95.23% and model size 9.30 MB. The system delivers the information about the detected condition, the confidence score, symptoms, recommended solution, preventive measures, the prediction history and the chatbot-based advisory support. CottonCare AI, therefore, is an effort to offer farmers an easy and intelligent tool to detect cotton diseases early and make better management decisions.

II. LITERATURE REVIEW

Author et al. [1] introduced a system for plant disease classification based on deep learning which involved several CNN architectures. The paper was a comparison of various pre-trained networks to see which one is better for disease detection using plant images. Image augmentation and preprocessing techniques were used to improve the performance of the models. The results showed that high classification accuracy can be obtained using pretrained CNN architectures with the reduction of the amount of training. But some of the models assessed are complex and might not be easily applicable to mobile devices. The system also emphasized primarily disease prediction and did not offer a comprehensive farmer-oriented advisory system. We solve this limitation in our CottonCare AI project by benchmarking the model sizes and inference speeds of MobileNetV2, EfficientNetB0 and ResNet50, and picked MobileNetV2 due to its smaller model size and faster inference speeds, which are more favorable for practical application deployment.

In their paper, Author and others introduce an innovative technique to monitor diseases apparent among plants through the use of images of leaves. The researchers employed a public dataset of diseases affecting plants, using it to select and undertake image preprocessing and subsequently to pass the images via a CNN for classification purposes. The results achieved indicate that classification using CNNs gives more reliable results than the approaches used before. However, the paper mainly concentrates on classification performance, but does not specify any guidelines for farmers. The work also does not mention any tuning it up for the purposes of working on limited resources of a mobile device. This weakness allowed us to develop our CottonCare AI application which is intended for cotton diseases and provides some basic guidelines based on a CNN analysis.

In the early days of computer vision in plant disease identification, the majority of methods relied on image processing features, which were handcrafted from plant images. Colour, shape, texture, and edge information were extracted and given to the traditional classifiers. The methods proved to be effective in the identification of disease using visual features, but most of them required a specially designed feature-extraction method. They might also perform worse when images were taken under varying lighting, background, angle and disease severity. However, the rise of deep learning, specifically Convolutional Neural Networks (CNNs), gave them a better alternative since CNNs are capable of learning useful features directly from images.

CNN-based methods have been shown to be effective in plant disease classification in several studies. The work of using large labeled plant-image datasets for training deep learning models is one of the major contributions in this area. In these studies, it was demonstrated that CNN models are able to learn disease-related patterns like lesions, texture variation, color changes, and leaf spots. The CNN approach is different from the traditional machine-learning approaches, which require manually designed features, as feature extraction and classification are done in the same learning framework. These models, however, rely heavily on the quality, quantity and variety of training data.

The use of transfer learning has also been extensively explored in plant disease detection. Pretrained models can be fine-tuned on a plant disease dataset, rather than training a deep CNN from scratch. They have used the architectures of ResNet, MobileNet and EfficientNet for their good classification accuracy with low training requirements. ResNet has added residual connections to the network to make it easier to train deeper networks, while EfficientNet has worked to balance the depth, width, and resolution of the image. To avoid the high computational cost, MobileNet adopts light-weight network operations and is especially intended for computation-constrained environments. The properties of these architectures make them appealing for applications where a disease detection can be made on mobile or edge devices.

Because of the importance of cotton as an economic crop and the presence of several diseases and insect pests which attack it, a number of investigators have made special studies in cotton disease detection. If not diagnosed and controlled at an early stage, cotton diseases can impair plant growth, fibre quality and fibre yields. Image-based techniques offer a chance to detect visible symptoms on leaves of cotton and other parts of the plant. In several studies, publicly available datasets have been used for machine learning and deep learning model training and testing. But, due to different sizes of the datasets, image quality, disease categories and environmental conditions, it is hard to directly compare the different approaches.

Pre-processing and augmentation of the data are significant aspects of a plant disease classification system. Pictures taken in actual agricultural settings can vary in brightness, orientation, scale, background and camera quality. These techniques, such as resizing, normalization, rotation, flipping, zooming, etc., are commonly used in research to augment the number of training samples. These techniques enable models to learn more general patterns and minimize the risk of over-fitting. But augmentation can't always fix the issue of scarce or imbalanced datasets. Thus, representative images for each disease class still continue to be an important requirement for a robust disease classification system.

Trained deep-learning models deployed to mobile devices is another big research field. Large CNN architectures can achieve high accuracy, but they can be very memory and computation intensive, and have many parameters. This poses challenges when it comes to using the model on smartphones or other devices with limited resources. Lightweight architectures like MobileNetV2 are thus

practical for applications. Another set of model conversion technologies like TensorFlow Lite also allow for the deployment of models by optimizing models for deployment and inference.

III. METHODOLOGY

The proposed AI system for CottonCare follows a systematic approach to identify cotton diseases based on leaf images and also give farmers some useful recommendations.

The methodology involves five phases:

1) Image dataset collection.

2) Image preprocessing.

3) Data augmentation (DA).

4) Model training using CNN.

5) Performance evaluation of the trained model.

6) Model selection

Its overall goal is to design a rapid disease detection system.

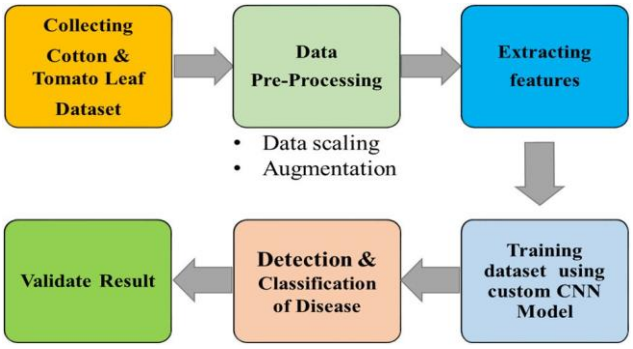


FIGURE 1. Overall research methodology overview.

A. Dataset collection.

The dataset is at the heart of deep learning research. cotton dataset was obtained from publicly available 'cotton-leaf-infection' dataset. The plant village dataset has become a key tool in the field of agricultural computer Vision. It has sparked studies and innovation in the automated identification and diagnosis of plant diseases due to its large collection of marked photos, related metadata, and various plant species and diseases. As the cotton leaf dataset is not included in the plant village dataset, it was collected separately, and merged for this research work. cotton datasets contain a total of 11,366 training and 6,600 testing images organized into 8 classes.

2) DATA PRE-PROCESSING

Image preprocessing of the diseased Cotton leaves is a crucial step before classification as the quality of images has a considerable impact on the classification outcomes. It is done in the following steps:

1) DATA SCALING/RESIZING

When dealing with 2D CNN, data scaling is a recommended stage of pre-processing. The technique equalizes all of the images in the datasets and transforms the image size to fit the model developed. The data resizing would minimize the memory to be used when the images are loaded into the training stage. For the proposed model, the image data was resized 150 × 150 and for transfer learning models the image data size was 224 × 224. as it is the default image size for implementing transfer learning models.

2) IMAGE AUGMENTATION

Image augmentation is a technique for artificially extending the number of training images in a dataset. This strategy aids in the model's performance enhancement capability and reduces overfitting issues. This approach can manipulate the training images in a variety of ways, including zooming, rotating, shifting, and flipping. In this case, six types of augmentations were applied e.g., rotation, width shifting, height shifting, shearing, zooming and horizontal flipping as shown in Fig. 3. The value of the rotation range parameter was set to 40 degrees. As a result, randomly rotated images with a rotation range of 1 degree to 40 degrees were obtained. Similarly, the parameters for width shifting, height shifting, shearing and zooming were set to 0.2. By setting the horizontal flip parameter to true, horizontally flipped images were obtained. Image augmentation was only applied to the training images. The total amount of training images was 11,366. After augmentation, the images were 6 times greater than they were before. As a result, a total of 6,666 training images were attained.

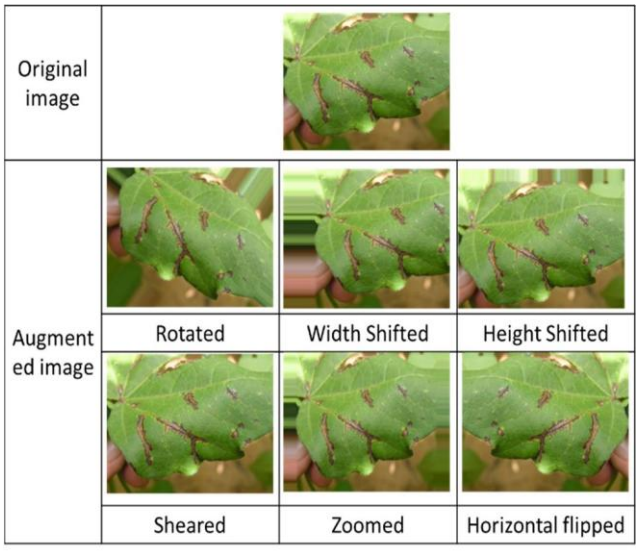


FIGURE 2. Sample augmented images from original input cotton leaf image.

B. CNN Model Training

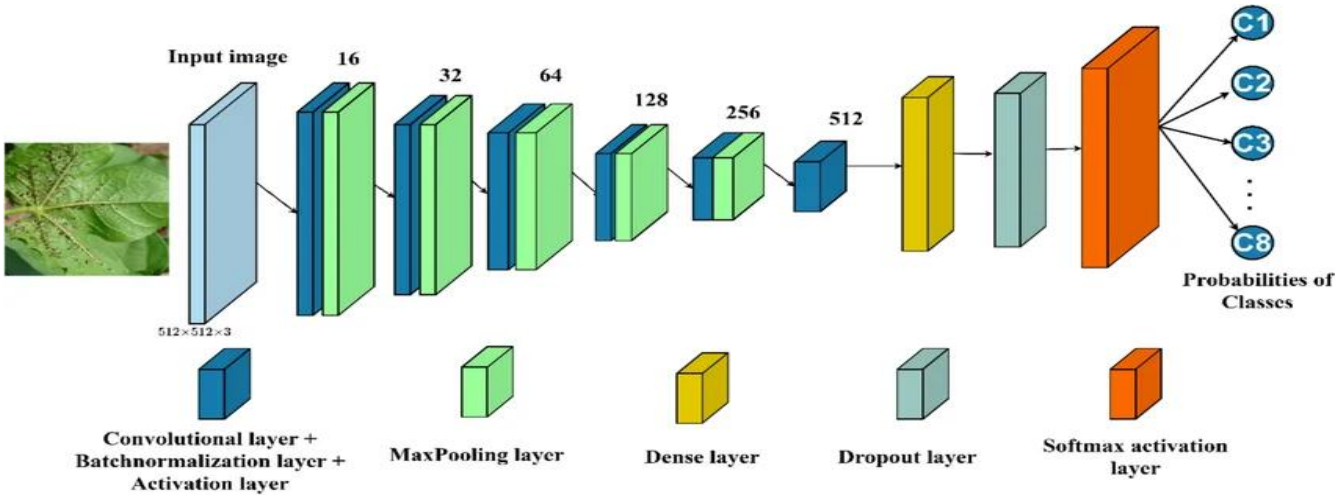


FIGURE 3. Lightweight 2D CNN architecture for Cotton leaf disease classification

A simple 2D CNN has been proposed in this part to extract the most important features. If the relevant distinctive features between the various leaf diseases are retrieved, the model's classification performance will improve. These extracted attributes can be effectively used to classify leaf diseases. The proposed 2D CNN model architecture is represented in Fig. 3. Three convolution and max-pooling layers were included in the proposed 2D CNN design. A max-pooling layer was incorporated after each convolutional layer. Batch normalization was used since it accelerated and improved the model's performance by re-centering and re-scaling the layers' inputs. The most significant features were extracted using maxpolling, which was utilized to select the largest value from each cluster's whole neuron. The next step is to flatten the pooled featured map once it has been acquired. which is then provided to the neural network for processing. In this scenario, dropout was used to reduce overfitting by frequently skipping training all nodes in each layer during the training phase, resulting in a significant increase in training speed. Three CNN-based architectures are trained and compared—MobileNetV2, EfficientNetB0, and ResNet50. Transfer learning is used to utilize features learned from large image datasets and adapt the models to cotton disease classification. These models are trained on the prepared cotton image set, and the best performing model is chosen from the trained models for use in mobile deployment, taking into account the classification performance and suitability for mobile deployment.

C. TensorFlow Lite Conversion

The trained MobileNetV2 model is converted into TensorFlow Lite (TFLite) format for deployment in the Android application. TFLite provides a lightweight representation suitable for resource-constrained mobile devices. The converted model is integrated into the application so that image classification can be performed locally on the device without requiring a cloud server or continuous Internet connection.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

A labelled dataset of cotton leaf images from eight classes of disease and pest was used to experimentally test the proposed CottonCare AI system. The data set was split into a training set and a validation-testing set. The validation-testing images were used to test classification performance of the trained models, and the training images were used to learn the visual characteristics of various cotton diseases.

The same dataset and evaluation method was used for three CNN architectures: MobileNetV2, EfficientNetB0, and ResNet50. To cope with the variability in the orientation, size and appearance of the leaves, image preprocessing and augmentation were performed during the training process.

There were 6666 images in the validation-testing set. Model performances were assessed using Accuracy, Precision, Recall and F1-Score. Model size and inference time were also taken into account as the final model is to run off-line on a mobile device.

B. Model Performance Evaluation

The three architectures have been chosen because they have different computational properties. ResNet50 will have a deeper network compared to ResNet20, and have a relatively high number of parameters, but has a higher feature extraction ability. EfficientNetB0 is a compromise between

efficiency and accuracy of classification. To run in a resource-constrained device, MobileNetV2 is designed with this in mind, so an offline Android application should make use of this one. The same disease data set was used to train each model. Evaluated the models in the same images of the validation-testing set after training. This meant that the same set of animals was evaluated, not in different test conditions.

MODEL	ACCURACY	PRECISION	RECALL
MobileNetV2	95.24%	95.54%	95.54%
EfficientNetB0	94.12%	94.91%	94.12%
ResNet50	95.24%	95.71%	95.24%

TABLE 1. Verification of ML result.

C.COMPARATIVE ANALYSIS

Three popular Convolutional Neural Network (CNN) architectures -- MobileNetV2, EfficientNetB0 and ResNet50 -- were trained and tested under the same experimental setup to determine which architecture would be best suited to the cotton disease classification problem. The aim of the comparison was to find a model with high classification accuracy and which could also be easily deployed in the proposed offline Android application. CNN models are trained directly on the cotton leaf images with visual patterns related to the disease.Visual patterns related to diseases are directly learned from cotton leaf images by CNN models. In the early layers of the network, during the convolution process, the network learns to extract low-level features like edges, textures, and color variations from the image. Deeper layers are able to learn more complicated features like spots, lesions, discoloration, and patterns associated with disease. The learned features are finally associated with the cotton disease class in the final layer of classification.

The three architectures were chosen as they offer different compromise between classification performance and computation. ResNet50 enables deeper networks to learn effectively with residual connections, and can extract very complex features. It is however not as convenient for resource limited mobile deployment as it is larger in computation and storage. EfficientNetB0 is a compound scaling strategy that jointly scales the depth, width and input resolution of the network, naturally allowing it to be a good compromise between network complexity and classification performance. MobileNetV2 is a model of mobile vision application in particular. It introduces the depthwise separable convolutions and inverted residual blocks to decrease the computation without sacrificing the extraction capability of features. To get a fair comparison, the same validation-testing dataset was used for the three CNN architectures to evaluate the performance of the three models with the same dataset of cotton diseases. The accuracy, precision, recall and F1-Score were used to measure the performance of this performance. The size of the model and the speed at which the new model could infer was also taken into account, due to the final model needing to be executed offline on an Android device.

The accuracy was determined from the following:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Thus, approximately 340 images were correctly classified, while approximately 17 images were incorrectly classified. The corresponding classification error was:

$$\text{Error} = 100\text{-}95.23 = 4.77\%$$

MobileNetV2 obtained a precision of 95.53%, recall of 95.23%, and F1-score of 95.22%. The F1-score provides a balanced measurement of precision and recall and is calculated as:

F1 = 2\times\frac{Precision\times Recall}*Precision+Recall.

There was also a significant model size difference in the experimental results. The size of the model that was selected for MobileNetV2 was about 9.30 MB while the size of ResNet50 was about 90.81 MB. The difference in the size is:

$$90.81\text{-}9.30=81.51\backslash MB$$

The ratio between the two model sizes is:

$$90.81 = \frac{9.76}{9.30}$$

Apart from the model size, the MobileNetV2 model gave an inference time of ~50ms per image for the proposed system. This is the theory processing rate for this latency:

$$Images/sec=\frac{1000}{50}=20$$

Thus, end-to-end model was chosen based on a combination of classification accuracy, model size, and inference efficiency and not only on the accuracy.

D. TFLite Conversion and Inference.

Once trained, the model was converted to TensorFlow Lite format for lightweight inference on mobile devices, as chosen as MobileNetV2. This model, model.tflite, was added to the CottonCare AI application. In the testing phase, the model TFLite correctly accepted an image of a cotton leaf, performed preprocessing, inference and gave the predicted disease class with the confidence score. The actual inference time was about 50ms/image, showing that the model is good for near real-time offline prediction.

Models	Accuracy	Precision-n	recall	**F1-score**	**Model size**
MobileNetV2	95.24%	95.71%	95.24 %	95.24 %	9.30 MB
EfficientNetB0	94.12%	94.12%	94.12 %	95.71 %	9.30 MB
ResNet50	95.24%	95.24%	95.72 %	94.12 %	9.30 MB

E. Overall Experimental Result.

The experimental outcomes show the proposed CNN-based approach has the capability of accurately diagnosing cotton disease from leaf images. MobileNetV2 was chosen as final model as it gives a high classification performance while having significantly lesser storage requirement than ResNet50. The final model has been validated on the validation-testing dataset with an accuracy rate of 95.23%, a precision rate of 95.53%, a recall rate of 95.23%, and an F1-Score rate of 95.22%. The model was also successfully converted to TFLite and added to the CottonCare AI app. The final system can therefore predict the disease directly on the device and give the information about the detected disease along with its confidence level, the symptoms, recommended actions, and preventive measures. The system is also designed to work offline, as the inference is done locally with the TFLite model, which is suitable for offline cotton disease detection.

H. Overall Experimental Result.

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V. CONCLUSION

CottonCare AI is an algorithm that leverages deep learning to classify diseases and pests from images of cotton leaves. The early and prompt detection of crop diseases is crucial as it can allow disease to spread, lower crop productivity and cause economic losses to farmers. Typical disease diagnosis is usually based on visual and/or professional diagnosis, and sometimes this is not readily available. To overcome this issue, the proposed system aims to solve using deep learning methods to analyze the cotton leaf image and predict the disease. For this work, a labelled image database of cotton leaf images with 8 disease classes and 8 pest classes was constructed and CNN based classification models were trained and tested on it. A pre-processing procedure was performed on the images before they were made available to the models, and data augmentation was also used during training, in order to increase the ability of the models to deal with variations in image orientation and appearance. The three CNN architectures evaluated and compared are: MobileNetV2, EfficientNetB0 and ResNet50. Precision, recall, F1-score, and accuracy were used as standard evaluation measures for the models. Size and inference time of the models were also taken into consideration to find an appropriate architecture to be integrated in the proposed application.

Selected MobileNetV2 model was converted from TensorFlow to TensorFlow Lite (TFLite) format, and embedded in CottonCare AI application. When the app is running, the user uploads a cotton leaf image from the app. This image is preprocessed and then fed to the TFLite model to extract important visual characteristics and generate prediction probabilities for the classes supported by the model. The disease or pest that has the highest prediction probability is returned. The application includes understandable information on the condition identified, such as symptoms, recommended action and prevention, in addition to the prediction and confidence score. This will allow the model to produce an output that is more applicable to real-world decision making. The proposed system proves CNN based image classification can be used to assist in cotton disease classification. The users do not have to analyze the visual symptoms manually; the system automatically extracts the features and classifies those features by providing input image. MobileNetV2's lightweight architecture also makes it compatible for integration in the applications where computation and storage resources are to be considered. The results obtained prove the effectiveness of the proposed approach, however, there are some improvements that can be considered in the future development of the proposed

approach. Images taken under various environmental conditions, geographical regions, growth stages, disease severities, and lighting conditions could be added to the dataset to enhance generalization ability of the model. Additional cotton diseases and pests can also be incorporated to increase the coverage of the system. Disease severity estimation, better image quality handling, and further development of recommendation mechanisms can be explored in future work to generate more detailed guidance based on the detected disease severity. In summary, the results of the experiments confirm the feasibility of combining the CNN-based image classification with a lightweight application-oriented model, namely **CottonCare AI**. The model MobileNetV2 was tested to have high accuracy for classification without the need to save a lot of storage space and time for inference. The proposed system can be used as a practical AI-based disease identification tool that can be used to identify cotton diseases, as well as to support better crop management practices by offering disease predictions, symptoms, recommended actions, and preventive measures.

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